

Supplementary Material to “Moments by Integrating the Moment-Generating Function”

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Abstract

This supplementary material contains auxiliary integral identities and oscillatory bounds, normal-inverse Gaussian parameter mappings, details on integrand behavior, comparisons with existing formulas, numerical implementation details, timing comparisons, and supplementary numerical checks for the parabolic representation.

Keywords: Fractional moments, moment-generating function, model-implied moments

S.1 Auxiliary Integral Identities and Oscillatory Bounds

S.1.1 The vanishing identity

Detailed Proof of Lemma A.1 Consider the function $f(t) = e^{-itx}w^{-(r+1)} = e^{-itx}(\sigma + it)^{-(r+1)}$. For $x > 0$, the term e^{-itx} decays exponentially in the lower half of the complex t -plane. We define a

closed contour C_R consisting of the interval $[-R, R]$ on the real axis and a semi-circle of radius R in the lower half-plane. Write $t = u + iv$ with $v \leq 0$ on C_R and set $w = \sigma + it$. The only potential

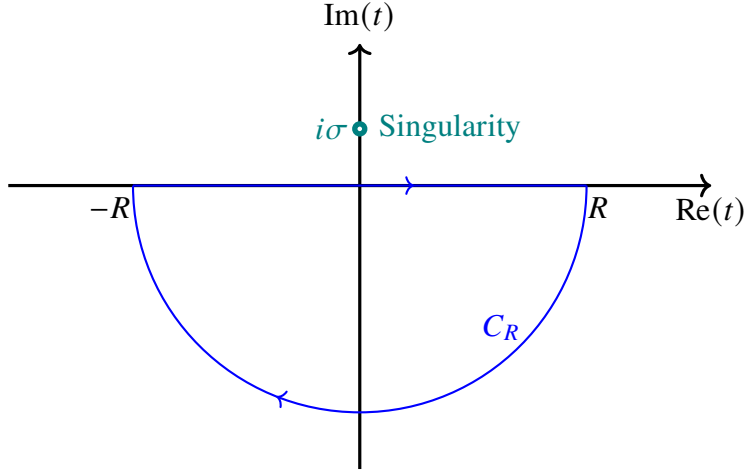


Figure S.1: The integration contour C_R in the complex t -plane. The path along the real axis $[-R, R]$ is closed by a semi-circle in the lower half-plane. The singularity at $t = i\sigma$ lies outside the contour.

singularity of the integrand arises at $w = 0$, equivalently at $t = i\sigma$. This point lies in the upper half-plane and is therefore outside C_R . Moreover, for $t = u + iv$ with $v \leq 0$ on (and inside) C_R we have $w = \sigma + it = \sigma + i(u + iv) = \sigma + iu - v$, $\text{Re}(w) = \sigma - v \geq \sigma > 0$, so the contour lies entirely in the half-plane $\text{Re}(w) > 0$ and hence avoids the principal branch cut of $\log(w)$ on $(-\infty, 0]$. Consequently, the integrand is analytic on and inside C_R , and by Cauchy's theorem, $\oint_{C_R} f(t)dt = 0$.

As $R \rightarrow \infty$, the integral along the circular arc vanishes due to the exponential decay of e^{-itx} (Jordan's Lemma) and the algebraic decay of $|w|^{-(r+1)}$. Therefore, the integral along the real axis must also be zero. $\square$

S.1.2 Oscillatory bounds

Lemma S.1 (Oscillatory bound). *Let $s > 0$, $z = s + it$, and let $p = \text{Re}(r) \in (-1, 0)$. For each fixed r there exists a constant $C_r < \infty$ such that, for every $x \neq 0$,*

$$\sup_{T>0} \left| \int_{-T}^T \frac{e^{zx} + e^{-zx}}{z^{r+1}} dt \right| \leq C_r (e^{sx} + e^{-sx}) |x|^p.$$

Proof. It is enough to consider one of the two exponential terms. For the term involving e^{zx} , factor

out the real exponential and write

$$\int_{-T}^T \frac{e^{(s+it)x}}{(s+it)^{r+1}} dt = e^{sx} I_T(x|r), \quad I_T(x|r) = \int_{-T}^T \frac{e^{itx}}{(s+it)^{r+1}} dt.$$

Set $u = t|x|$, $S = T|x|$, and $a = s|x|$. Then

$$I_T(x|r) = |x|^r \int_{-S}^S \frac{e^{iu \operatorname{sgn}(x)}}{(a+iu)^{r+1}} du.$$

It is therefore sufficient to show that, for a constant depending only on r ,

$$\sup_{S>0, a>0} \left| \int_0^S \frac{e^{\pm iu}}{(a \pm iu)^{r+1}} du \right| < \infty.$$

Let $U > 0$ be fixed. On the interval $[0, U]$, we have

$$\left| \int_0^U \frac{e^{\pm iu}}{(a \pm iu)^{r+1}} du \right| \leq e^{\pi |\operatorname{Im}(r)|} \int_0^U u^{-p-1} du = \frac{e^{\pi |\operatorname{Im}(r)|} U^{-p}}{-p},$$

which is finite because $p < 0$.

For the interval $[U, S]$, assuming $S > U$, integration by parts gives

$$\int_U^S \frac{e^{\pm iu}}{(a \pm iu)^{r+1}} du = \left[\pm \frac{e^{\pm iu}}{i(a \pm iu)^{r+1}} \right]_U^S + (r+1) \int_U^S \frac{e^{\pm iu}}{(a \pm iu)^{r+2}} du.$$

The remainder is bounded by

$$\int_U^S \left| \frac{1}{(a \pm iu)^{r+2}} \right| du \leq e^{\pi |\operatorname{Im}(r)|} \int_U^S u^{-p-2} du \leq \frac{e^{\pi |\operatorname{Im}(r)|}}{p+1} U^{-p-1},$$

because $p > -1$. The boundary terms are also bounded by a constant depending only on r and U .

Hence

$$\sup_{S>0, a>0} \left| \int_0^S \frac{e^{\pm iu}}{(a \pm iu)^{r+1}} du \right| \leq C_r < \infty.$$

Consequently,

$$|I_T(x|r)| \leq C_r |x|^p.$$

Multiplying by the outer factor e^{sx} gives the bound for the term involving e^{zx} . The term involving

e^{-zx} is obtained by replacing x with $-x$, and gives the factor e^{-sx} . Combining the two terms yields

$$\sup_{T>0} \left| \int_{-T}^T \frac{e^{zx} + e^{-zx}}{z^{r+1}} dt \right| \leq C_r (e^{sx} + e^{-sx}) |x|^p.$$

This completes the proof. $\square$

Lemma S.2 (Boundary oscillatory bound). *Let $s > 0$, $z = s + it$, and let $r = i\eta$ with $\eta \in \mathbb{R}$. For each fixed η there exists a constant $C_\eta < \infty$ such that, for every $x \neq 0$,*

$$\sup_{T>0} \left| \int_{-T}^T \frac{e^{zx} + e^{-zx}}{z^{1+i\eta}} dt \right| \leq C_\eta (e^{sx} + e^{-sx}).$$

For $\eta = 0$, the same conclusion holds with $C_0 < \infty$.

Proof. It is enough to bound one exponential term. Fix $\varepsilon \in \{-1, 1\}$ and consider

$$K_{\varepsilon,T}(x) = \int_{-T}^T \frac{e^{\varepsilon zx}}{z^{1+i\eta}} dt.$$

Set $u = t|x|$, $S = T|x|$, and $a = s|x|$. Since $z = s + it = (a + iu)/|x|$, we obtain

$$K_{\varepsilon,T}(x) = e^{\varepsilon sx} |x|^{i\eta} \int_{-S}^S \frac{e^{i\varepsilon u \operatorname{sgn}(x)}}{(a + iu)^{1+i\eta}} du.$$

Because $|x|^{i\eta}$ has modulus one, it remains to show that

$$\sup_{S>0, a>0} \left| \int_{-S}^S \frac{e^{i\varepsilon u}}{(a + iu)^{1+i\eta}} du \right| < \infty,$$

with a constant depending only on η .

Fix $U > 0$ and split the integral into the central part $|u| \leq U$ and the tails $|u| > U$. For the central part, write

$$e^{i\varepsilon u} = 1 + \{e^{i\varepsilon u} - 1\}.$$

First suppose $\eta \neq 0$. Since

$$\frac{d}{du} (a + iu)^{-i\eta} = \eta (a + iu)^{-1-i\eta},$$

we have, for $0 < B \leq U$,

$$\left| \int_{-B}^B (a + iu)^{-1-i\eta} du \right| \leq \frac{2e^{|\eta|\pi/2}}{|\eta|}.$$

For $\eta = 0$,

$$\int_{-B}^B \frac{1}{a + iu} du = \frac{1}{i} \{ \log(a + iB) - \log(a - iB) \},$$

and this is bounded uniformly in $a > 0$ and $B > 0$. The remaining central term is also uniformly bounded, since $|e^{i\epsilon u} - 1| \leq |u|$ and

$$\int_{-B}^B \frac{|u|}{|a + iu|} du \leq 2B \leq 2U,$$

up to the harmless phase factor coming from $(a + iu)^{-i\eta}$. Thus the contribution from $|u| \leq U$ is bounded uniformly in S and a .

It remains to bound the tails. On $[U, S]$, integration by parts gives

$$\int_U^S \frac{e^{i\epsilon u}}{(a + iu)^{1+i\eta}} du = \left[\frac{e^{i\epsilon u}}{i\epsilon(a + iu)^{1+i\eta}} \right]_U^S + \frac{1 + i\eta}{\epsilon} \int_U^S \frac{e^{i\epsilon u}}{(a + iu)^{2+i\eta}} du.$$

Since $|a + iu| \geq u$ for $u > 0$, the boundary terms are bounded by a constant times U^{-1} , and the remainder is bounded by a constant times

$$\int_U^\infty u^{-2} du < \infty.$$

The same argument applies on $[-S, -U]$. Therefore

$$\sup_{S>0, a>0} \left| \int_{-S}^S \frac{e^{i\epsilon u}}{(a + iu)^{1+i\eta}} du \right| \leq C_\eta < \infty.$$

Consequently,

$$\sup_{T>0} \left| \int_{-T}^T \frac{e^{zx}}{z^{1+i\eta}} dt \right| \leq C_\eta e^{sx}, \quad \sup_{T>0} \left| \int_{-T}^T \frac{e^{-zx}}{z^{1+i\eta}} dt \right| \leq C_\eta e^{-sx}.$$

Adding the two bounds gives

$$\sup_{T>0} \left| \int_{-T}^T \frac{e^{zx} + e^{-zx}}{z^{1+i\eta}} dt \right| \leq C_\eta (e^{sx} + e^{-sx}).$$

The proof includes the case $\eta = 0$ by the separate central bound above. $\square$

The proofs of Lemmas S.1 and S.2 also give the corresponding one-sided estimates

$$\sup_{T>0} \left| \int_{-T}^T \frac{e^{zx}}{z^{r+1}} dt \right| \leq C_r e^{sx} |x|^{\operatorname{Re}(r)}, \quad -1 < \operatorname{Re}(r) < 0,$$

and

$$\sup_{T>0} \left| \int_{-T}^T \frac{e^{zx}}{z^{1+i\eta}} dt \right| \leq C_\eta e^{sx}, \quad \eta \in \mathbb{R}.$$

Lemma S.3 (Logarithmic oscillatory bound). *Let $s > 0$, $z = s + it$, and let $p = \operatorname{Re}(r) \in (-1, 0]$.*

For each fixed r there exists a constant $C_{r,s} < \infty$ such that, for every $x \neq 0$,

$$\sup_{T>0} \left| \int_{-T}^T \frac{e^{zx} + e^{-zx}}{z^{r+1}} \log(z) dt \right| \leq C_{r,s} (e^{sx} + e^{-sx}) |x|^p (1 + |\log |x||),$$

where $\log(z)$ denotes the principal logarithm.

Proof. It is enough to consider one of the two exponential terms. For the term involving e^{zx} , write

$$\int_{-T}^T \frac{e^{zx}}{z^{r+1}} \log(z) dt = e^{sx} \int_{-T}^T \frac{e^{itx}}{(s + it)^{r+1}} \log(s + it) dt.$$

Set $u = t|x|$, $S = T|x|$, and $a = s|x|$. Since $s + it = (a + iu)/|x|$, we have

$$\log(s + it) = \log(a + iu) - \log |x|,$$

and therefore

$$\int_{-T}^T \frac{e^{itx}}{(s + it)^{r+1}} \log(s + it) dt = |x|^r \int_{-S}^S \frac{e^{iu \operatorname{sgn}(x)}}{(a + iu)^{r+1}} \{\log(a + iu) - \log |x|\} du.$$

First suppose $p \in (-1, 0)$. The term involving $-\log |x|$ is controlled by Lemma S.1 and contributes at most a constant times

$$|x|^p |\log |x||.$$

It remains to bound the term with $\log(a + iu)$. The same splitting and integration-by-parts argument used in Lemma S.1 applies, with the logarithmic factor carried through the estimates. On bounded

intervals, the integrand is bounded by an integrable envelope of the form

$$|u|^{-p-1}(1 + |\log |u||),$$

away from $u = 0$, while the region near $u = 0$ contributes at most a constant times $1 + |\log a|$. Since $a = s|x|$, this term is bounded by a constant times $1 + |\log |x||$, with the constant allowed to depend on s . On the tails, integration by parts gives boundary terms of order

$$(1 + \log u)u^{-p-1}$$

and a remainder bounded by

$$\int_U^\infty (1 + \log u)u^{-p-2}du < \infty, \quad p > -1.$$

Thus, for $p \in (-1, 0)$, the logarithmic term is bounded by

$$C_{r,s}|x|^p(1 + |\log |x||).$$

Now consider the boundary case $p = 0$, so $r = i\eta$ with $\eta \in \mathbb{R}$. In the decomposition above, the term involving $-\log |x|$ is controlled by Lemma S.2 and contributes at most a constant times $|\log |x||$. It remains to bound the term with $\log(a + iu)$.

Fix $\varepsilon \in \{-1, 1\}$ and write

$$J_{\varepsilon,S,a} = \int_{-S}^S \frac{e^{i\varepsilon u}}{(a + iu)^{1+i\eta}} \log(a + iu) du.$$

We show that

$$\sup_{S>0} |J_{\varepsilon,S,a}| \leq C_{\eta,s}(1 + |\log a|).$$

Since $a = s|x|$, this gives the desired factor $1 + |\log |x||$.

Fix $U = 1$ and first consider the central part $|u| \leq U$. Write $e^{i\varepsilon u} = 1 + \{e^{i\varepsilon u} - 1\}$. For the term

with 1, if $\eta \neq 0$, an antiderivative is

$$\frac{1}{\eta}(a + iu)^{-i\eta} \left\{ \log(a + iu) + \frac{1}{i\eta} \right\}.$$

Thus the integral over any interval $[-B, B]$, $0 < B \leq U$, is bounded by $C_\eta(1 + |\log a|)$. If $\eta = 0$, the corresponding antiderivative is $\frac{1}{2i}\{\log(a + iu)\}^2$. Evaluating between $-B$ and B again gives a bound of order $C(1 + |\log a|)$, because the imaginary part of $\log(a + iB)$ is uniformly bounded. For the remaining central term, $|e^{i\epsilon u} - 1| \leq |u|$, and

$$\int_{-U}^U \frac{|u|}{|a + iu|} |\log(a + iu)| du \leq C(1 + |\log a|).$$

Hence the central part is bounded by $C_\eta(1 + |\log a|)$.

It remains to bound the tails. On $[U, S]$, integration by parts gives

$$\int_U^S \frac{e^{i\epsilon u} \log(a + iu)}{(a + iu)^{1+i\eta}} du = \left[\frac{e^{i\epsilon u} \log(a + iu)}{i\epsilon(a + iu)^{1+i\eta}} \right]_U^S - \frac{1}{i\epsilon} \int_U^S e^{i\epsilon u} \frac{d}{du} \left\{ \frac{\log(a + iu)}{(a + iu)^{1+i\eta}} \right\} du.$$

The boundary terms are uniformly bounded in S and a , since $|a + iu| \geq U$ and $(1 + |\log v|)/v$ is bounded for $v \geq U$. Moreover,

$$\left| \frac{d}{du} \left\{ \frac{\log(a + iu)}{(a + iu)^{1+i\eta}} \right\} \right| \leq C_\eta \frac{1 + |\log(a + iu)|}{|a + iu|^2}.$$

The integral of the right-hand side over $[U, \infty)$ is bounded uniformly in $a > 0$. To see this, if $a \leq U$ then $|a + iu| \geq u$ and the envelope is bounded by $C(1 + \log u)u^{-2}$. If $a > U$, split the integral into $[U, a]$ and $[a, \infty)$; the first part is bounded by $C(1 + \log a)/a$, and the second by $C \int_a^\infty (1 + \log u)u^{-2} du$. Both are uniformly bounded. The same argument applies on the negative tail $[-S, -U]$.

Combining the central and tail bounds gives

$$\sup_{s>0} |J_{\epsilon, S, a}| \leq C_{\eta, s}(1 + |\log a|) \leq C_{\eta, s}(1 + |\log |x||).$$

Multiplying by the outer factor e^{sx} gives the desired bound for the term with e^{zx} . The term with

e^{-zx} follows by replacing x with $-x$. □

S.2 Standardized NIG Parameter Mappings

For the numerical illustrations we utilize standardized normal-inverse Gaussian (NIG) distributions (zero mean and unit variance). Following [Barndorff-Nielsen et al. \(1985\)](#), these can be characterized entirely by two shape parameters, (ξ, χ) , satisfying $0 \leq |\chi| < \xi < 1$. The exact algebraic mapping to the standard parameters $(\alpha, \beta, \mu, \delta)$ is given by:

$$\alpha = \xi\zeta, \quad \beta = \chi\zeta, \quad \mu = -\frac{\chi(\xi^2 - \chi^2)}{\xi^2}\zeta, \quad \delta = \frac{(\xi^2 - \chi^2)^{3/2}}{\xi^2}\zeta, \quad \text{where } \zeta = \frac{\sqrt{1 - \xi^2}}{\xi^2 - \chi^2}.$$

For the absolute-moment formula in Theorem 1, the shift must be such that both $M_X(s + it)$ and $M_X(-s - it)$ are finite. Hence the relevant condition is $0 < s < \alpha - |\beta|$. For the two standardized distributions considered in the main text, these upper limits are $3\sqrt{3}/5 \approx 1.04$ and $2\sqrt{7} \approx 5.29$, respectively, so the choice $s = 1$ is admissible in both cases.

S.3 Integrand Behavior for the NIG Distribution

The shape of the integrands provides visual intuition for the computational stability of the CMGF method relative to classical density integration. Figure [S.2](#) plots the corresponding integrands for the NIG distribution with $(\xi, \chi) = (1/2, -1/3)$.

The CMGF integrand is very stable: it is non-negligible over a nearly identical domain for all values of r , and the range of the integrand is completely consistent across r . In fact, all CMGF integrands share the exact same maximum value at zero. By contrast, the classical integrand, $|x|^r f(x)$, exhibits severe structural variation across different moments, explaining why classical quadrature methods struggle and slow down significantly for higher fractional powers.

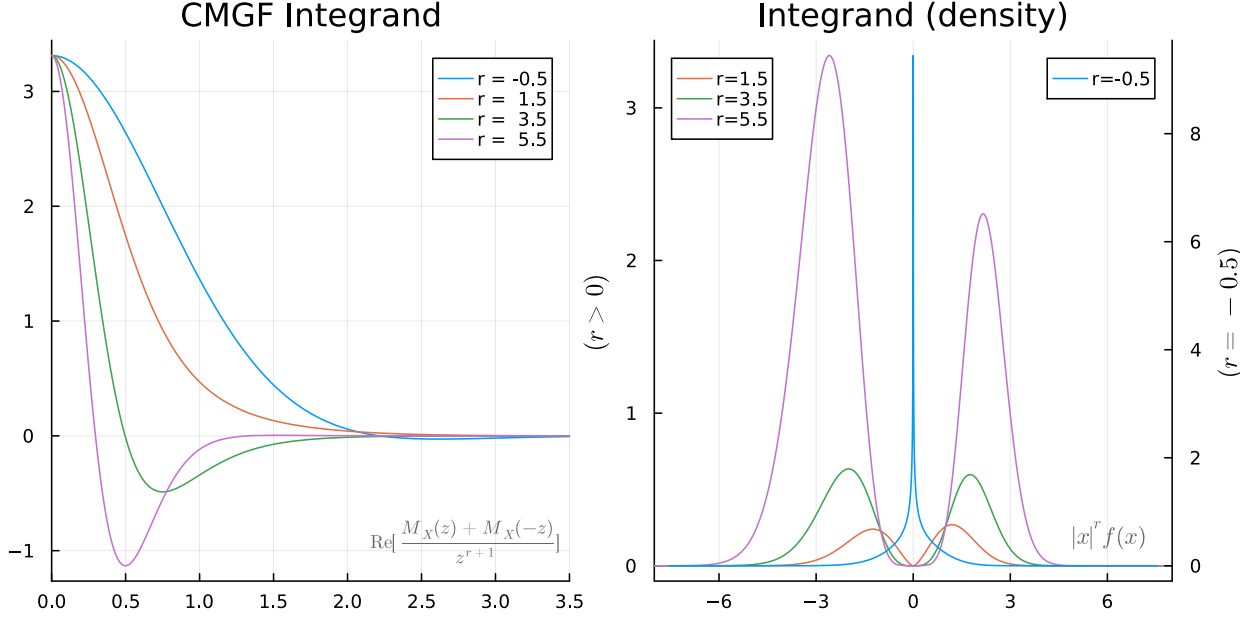


Figure S.2: Integrands for selected moments of the NIG distribution. The CMGF integrands (left) remain stable across fractional powers r , whereas the density-based integrands (right) vary substantially in scale and effective domain.

S.4 Existing Integral Expressions for Fractional Moments

To contextualize the theoretical contribution of the CMGF framework, we summarize the existing integral expressions for fractional moments, which predominantly rely on characteristic functions (CF) or fractional derivatives rather than the complex MGF.

For positive fractional moments, (Kawata, 1972, Theorem 11.4.4) established:

$$\mathbb{E}|X|^r = C_K \int_0^\infty u^{-(1+r)} \left[-\text{Re}[\varphi_X(u)] + \sum_{k=0}^K \frac{u^{2k}}{(2k)!} \varphi_X^{(2k)}(0) \right] du,$$

where $K = \lfloor r/2 \rfloor$ and C_K is a positive constant. Similarly, Laue (1980) derived:

$$\mathbb{E}|X|^r = \frac{\lambda}{\Gamma(1-\lambda)} [\cos(\frac{\pi r}{2})]^{-1} \int_0^\infty \frac{\varphi_X^{(k)}(0) - \varphi_X^{(k)}(u)}{u^{1+\lambda}} du,$$

where $k = \lfloor r \rfloor$ and $\lambda = r - k$.

For strictly non-negative random variables ($X \geq 0$), Schürger (2002) provided an expression

closely related to Laue's:

$$\mathbb{E}[X^r] = (-1)^k \frac{\lambda}{\Gamma(1-\lambda)} \int_0^\infty \frac{M_X^{(k)}(0) - M_X^{(k)}(-u)}{u^{1+\lambda}} du.$$

An alternative expression for non-negative variables utilizing higher-order derivatives was established by [Cressie and Borkent \(1986\)](#):

$$\mathbb{E}[X^r] = \frac{1}{\Gamma(\tilde{\lambda})} \int_0^\infty u^{\tilde{\lambda}-1} M_X^{(\tilde{k})}(-u) du,$$

where $\tilde{k} = \lceil r \rceil > 0$ and $\tilde{\lambda} = \tilde{k} - r \in [0, 1)$. Finally, for strictly positive variables ($X > 0$) and negative fractional moments ($r < 0$), ([Schürger, 2002](#), Theorem 1.1) established:

$$\mathbb{E}[X^r] = \frac{1}{\Gamma(-r)} \int_0^\infty \frac{1}{t^{r+1}} M_X(-t) dt.$$

The CMGF framework introduced in this paper unifies these previously separate strands in the sense that the same Fourier-Laplace mechanism yields derivative-free formulas for absolute moments, positive-part moments, signed integer moments, logarithmic moments, complex powers, and negative moments under explicit regularity conditions.

S.5 Numerical Implementation and Supplementary Checks

This section provides the computational details for the numerical illustrations, the timing comparison for the NIG example, and a supplementary numerical check of the parabolic representation in Theorem 4.

S.5.1 Hardware and Software Environment

All numerical evaluations, integrations, and Monte Carlo simulations were executed single-threaded using the Julia programming language, version 1.12.6 ([Bezanson et al., 2017](#)), on macOS (arm64-apple-darwin24). Computation times were evaluated using the `BenchmarkTools.jl` package of [Chen and Revels \(2016\)](#) (v1.8.0), which executes multiple samples to provide robust

minimum, median, and mean execution times. The algorithms were run on an Apple desktop computer equipped with an M2 Max processor and 64 GB of unified memory. The key packages used were `Distributions.jl` (v0.25.127), `QuadGK.jl` (v2.11.3), `SpecialFunctions.jl` (v2.8.0), `ForwardDiff.jl` (v1.4.1), and `Plots.jl` (v1.41.6).

S.5.2 Numerical Integration Tolerances

To ensure a fair comparison in Table 1, both the CMGF integral (evaluated via Theorem 1) and the classical density-based expectation, $\int_{-\infty}^{\infty} |x|^r f(x) dx$, were computed using the same adaptive numerical quadrature routines in Julia. Both methods were subject to identical relative and absolute error tolerance thresholds (e.g., `rtol` = `1e-10`). The computational stability of the CMGF integral is highly insensitive to the specific choice of s , provided it falls strictly within the convergence strip.

S.5.3 Monte Carlo Simulation Setup

The simulation-based moment estimates ($\hat{\mu}_N^r$) reported in Section 3 were computed as the empirical average of $N = 1,000,000$ independent and identically distributed random draws from the respective NIG distributions:

$$\hat{\mu}_N^r = \frac{1}{N} \sum_{j=1}^N |X_j|^r.$$

To accurately quantify the simulation error and determine the number of reliable decimal places plotted in Figure 2, we estimated the variance of the fractional powers, $\sigma^2(r) = \text{var}(|X_j|^r)$, using an extended auxiliary simulation of 100,000,000 draws. The number of accurate decimal places corresponding to a one-standard-deviation error was then calculated as $-\log_{10}(\sigma(r)/\sqrt{N})$.

S.5.4 Numerical check of the parabolic representation

As a numerical check of Theorem 4, we consider $X \sim \chi_8^2$. Its MGF is $M_X(z) = (1 - 2z)^{-4}$ for $\text{Re}(z) < 1/2$, and its fractional moments are available in closed form:

$$\mathbb{E}[X^r] = 2^r \frac{\Gamma(4+r)}{\Gamma(4)}, \quad r > -4.$$

Using the parabolic path $z = \sqrt{s} + it$ with $s = 0.1$, the integral in Theorem 4 matches the closed-form expression to machine precision over a wide range of positive and negative powers. This includes negative fractional moments close to the existence boundary $r = -4$.

We also evaluate the pole cases $r = -1, -2, -3$. Although the formula in Theorem 4 contains the factor $\Gamma(r + 1)$, the right-hand side is understood in the limiting sense at $r = -1, -2, \dots$. The rows labeled “limit” in Table S.1 evaluate the theorem slightly away from the poles, at $r = -m + 10^{-6}$. The rows labeled “log” use the pole-free logarithmic integral formula from Corollary 3 at the exact negative integers. For $X \sim \chi_8^2$, the corresponding moments are $\mathbb{E}[X^{-1}] = 1/6$, $\mathbb{E}[X^{-2}] = 1/24$, and $\mathbb{E}[X^{-3}] = 1/48$.

Table S.1: Numerical verification of the parabolic representation for $X \sim \chi_8^2$ with $s = 0.1$.

r	Method	Parabolic integral	Exact moment	Relative error
-3.5	limit	$2.611\,071\,119\,407\,293 \times 10^{-2}$	$2.611\,071\,119\,407\,292 \times 10^{-2}$	1.329×10^{-16}
-3	log	$2.083\,333\,333\,333\,322 \times 10^{-2}$	$2.083\,333\,333\,333\,333 \times 10^{-2}$	5.329×10^{-15}
$-3 + 10^{-6}$	limit	$2.083\,333\,574\,873\,437 \times 10^{-2}$	$2.083\,333\,574\,859\,052 \times 10^{-2}$	6.905×10^{-12}
-2.5	limit	$2.611\,071\,119\,407\,292 \times 10^{-2}$	$2.611\,071\,119\,407\,292 \times 10^{-2}$	0.000
-2	log	$4.166\,666\,666\,666\,709 \times 10^{-2}$	$4.166\,666\,666\,666\,666 \times 10^{-2}$	1.033×10^{-14}
$-2 + 10^{-6}$	limit	$4.166\,671\,316\,371\,828 \times 10^{-2}$	$4.166\,671\,316\,385\,254 \times 10^{-2}$	3.222×10^{-12}
-1.5	limit	$7.833\,213\,358\,221\,879 \times 10^{-2}$	$7.833\,213\,358\,221\,877 \times 10^{-2}$	1.772×10^{-16}
-1	log	$1.666\,666\,666\,666\,667 \times 10^{-1}$	$1.666\,666\,666\,666\,667 \times 10^{-1}$	1.665×10^{-16}
$-1 + 10^{-6}$	limit	$1.666\,669\,359\,805\,728 \times 10^{-1}$	$1.666\,669\,359\,888\,365 \times 10^{-1}$	4.958×10^{-11}
-0.5	limit	$3.916\,606\,679\,110\,938 \times 10^{-1}$	$3.916\,606\,679\,110\,939 \times 10^{-1}$	2.835×10^{-16}
0.5	limit	2.741 624 675 377 657	2.741 624 675 377 657	1.620×10^{-16}
1	limit	8.000 000 000 000 002	8.000 000 000 000 000	2.220×10^{-16}
2	limit	$8.000\,000\,000\,000\,003 \times 10^1$	$8.000\,000\,000\,000\,000 \times 10^1$	3.553×10^{-16}
3	limit	$9.599\,999\,999\,999\,997 \times 10^2$	$9.600\,000\,000\,000\,000 \times 10^2$	3.553×10^{-16}

Note: The numerical results use $s = 0.1$, safely inside the admissible range $0 < s < 1/2$. The rows labeled “limit” use Theorem 4; near the poles of $\Gamma(r + 1)$ they evaluate the integral at $r = -m + 10^{-6}$. The rows labeled “log” use the pole-free logarithmic integral formula of Corollary 3 at the exact negative integers. Values of s closer to the boundary remain valid, but can be less stable near the poles of $\Gamma(r + 1)$, where the limiting formula involves cancellation.

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